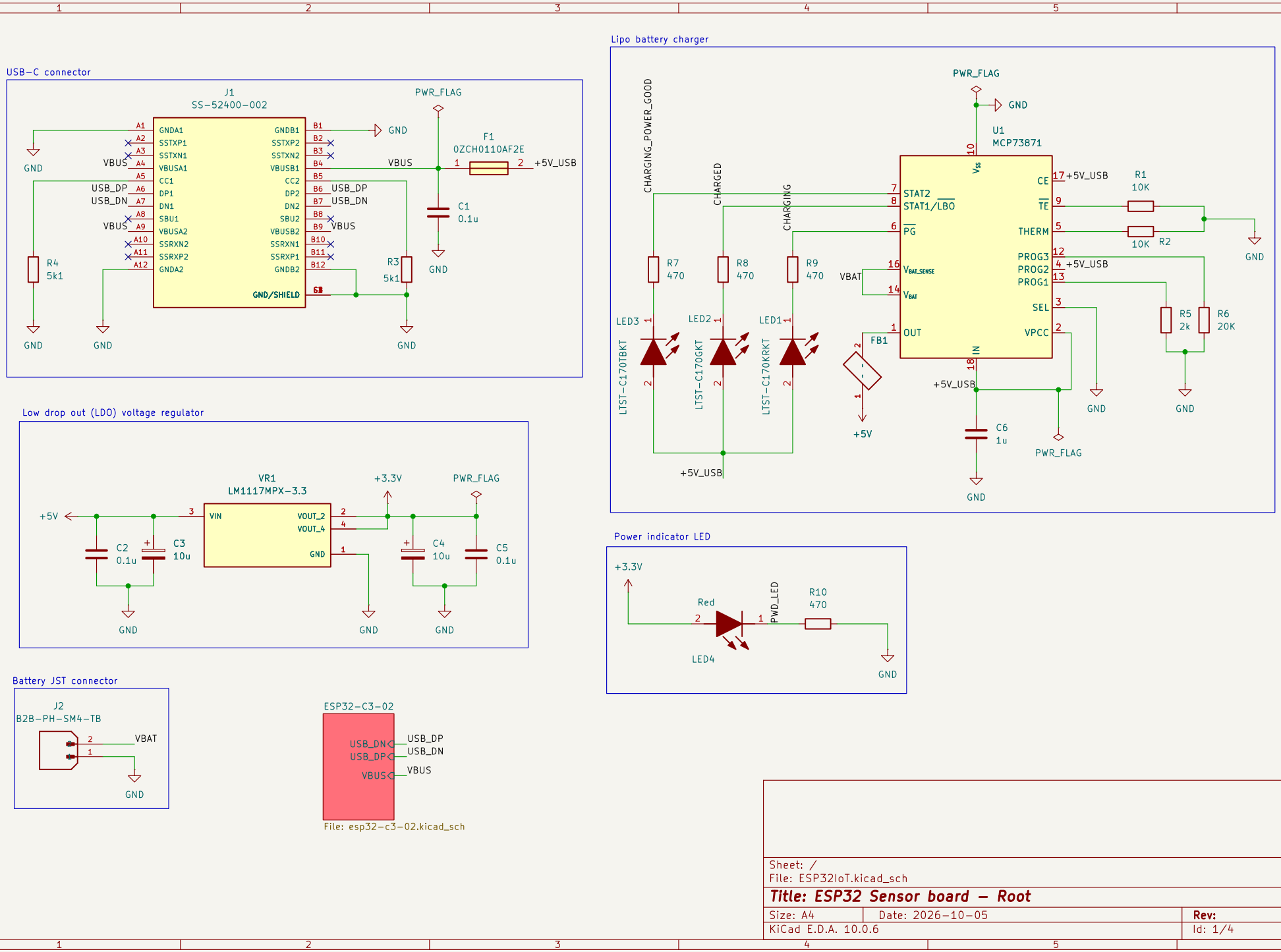
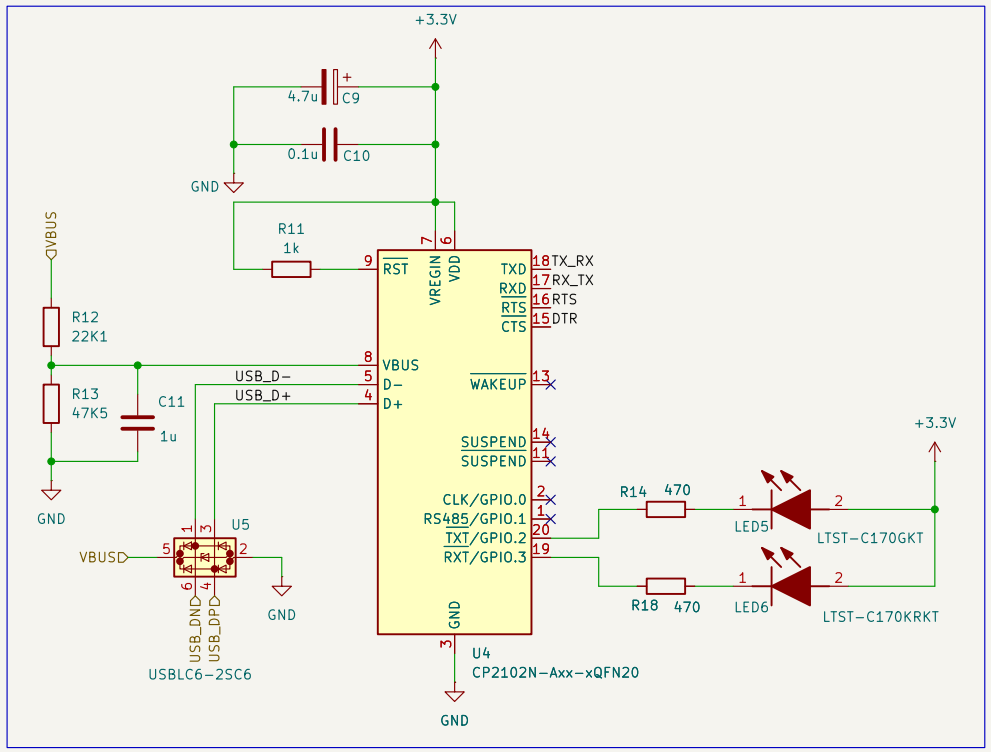
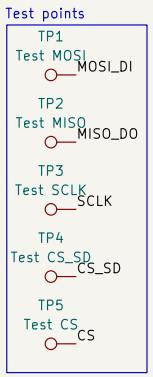
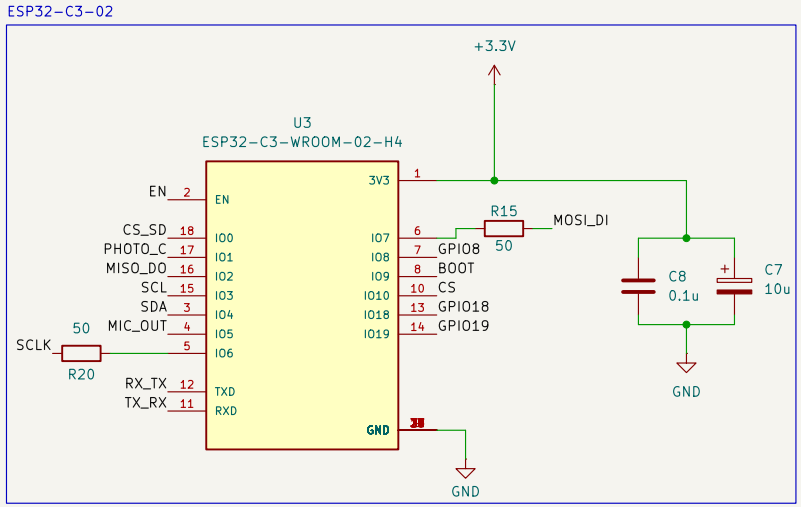
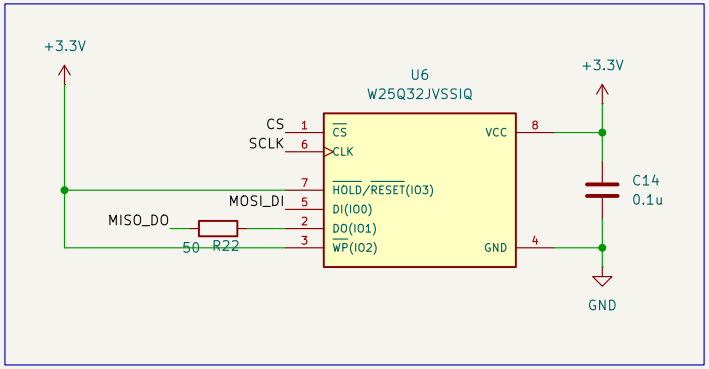
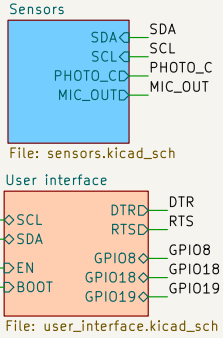

USB-C connector
J1
SS-52400-002
GNDA1
SSTXP1
SSTXN1
VBUSA1
CC1
DP1
DN1
SBU1
VBUSA2
SSRXN2
SSRXP2
GNDA2
GNDB1
SSTXP2
SSTXN2
VBUSB1
CC2
DP2
DN2
SBU2
VBUSB2
SSRXN1
SSRXP1
GNDB2
GND/SHIELD
PWR_FLAG
F1
0ZCH0110AF2E
+5V_USB
C1
0.1u
R3
5k1
R4
5k1
USB_DP
USB_DN
VBUS
GND
Lipo battery charger
U1
MCP73871
CHARGING_POWER_GOOD
CHARGED
CHARGING
STAT2
STAT1/LBO
PG
VBAT_SENSE
VBAT
OUT
FB1
+5V
Vss
CE
TE
THERM
PROG3
PROG2
PROG1
SEL
VPCC
IN
R1
10K
R2
10K
R5
2k
R6
20K
R7
470
R8
470
R9
470
LED1
LED2
LED3
LTST-C170KRKT
LTST-C170GKT
LTST-C170TBKT
C6
1u
Low drop out (LDO) voltage regulator
VR1
LM1117MPX-3.3
VIN
VOUT_2
VOUT_4
+5V
+3.3V
C2
0.1u
C3
10u
C4
10u
C5
0.1u
Power indicator LED
Red
PWD_LED
R10
470
LED4
Battery JST connector
J2
B2B-PH-SM4-TB
VBAT
ESP32-C3-02
USB_DN
USB_DP
VBUS
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Title: ESP32 Sensor board - Root
Size: A4
Date: 2026-10-05
Rev:
KiCad E.D.A. 10.0.6
Id: 1/4

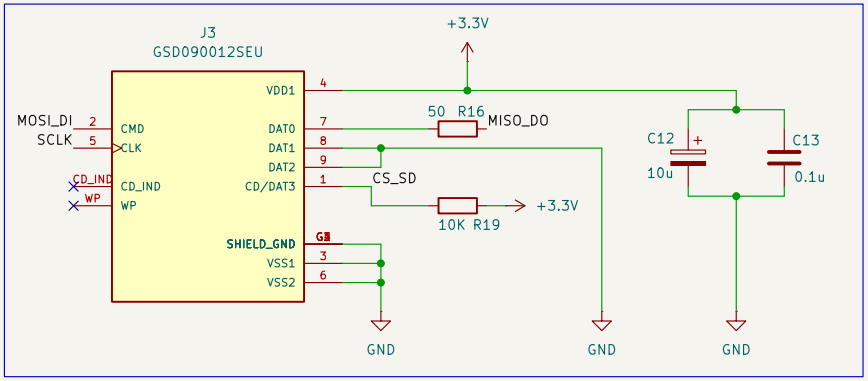












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Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 2/4

BME280

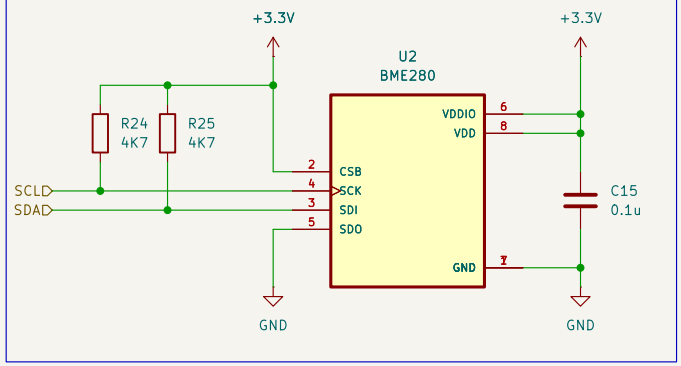
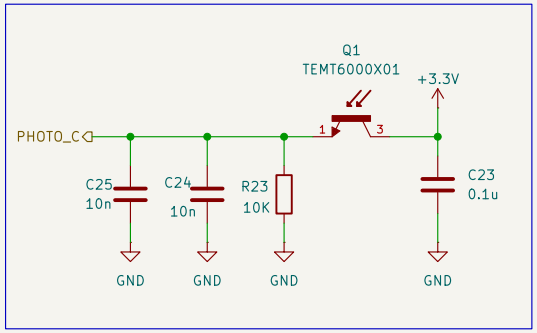
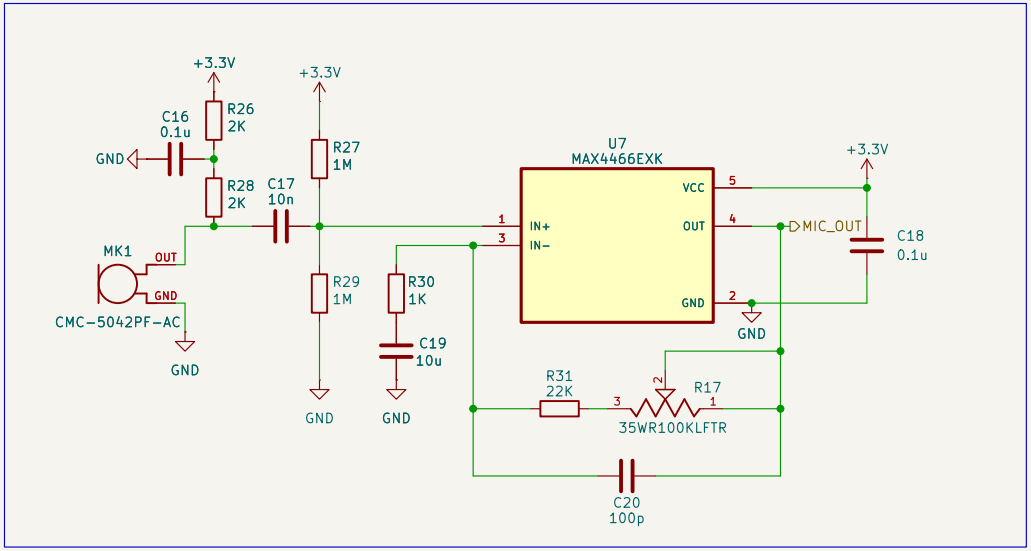


PHOTO SENSOR



Microphone and amplifier

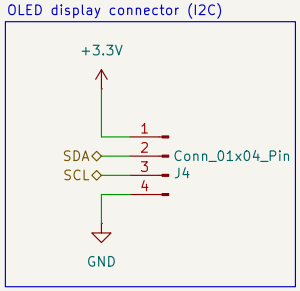


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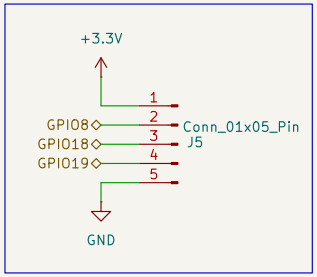
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KiCad E.D.A. 10.0.6		Id: 3/4

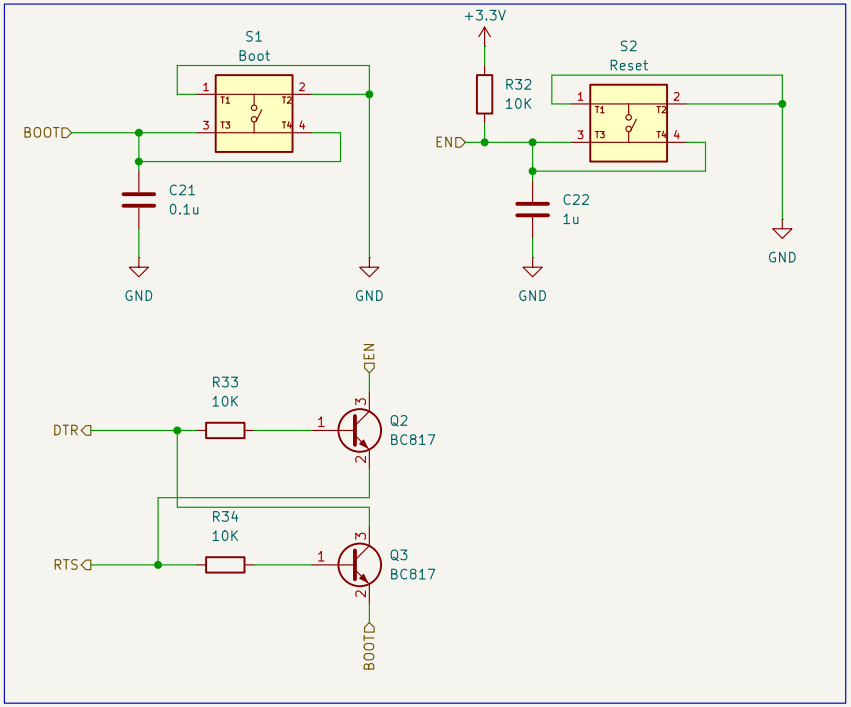
OLED display connector (I2C)



Breakout for remaining pins



RESET and BOOT buttons



Sheet: /ESP32-C3-02/User interface/		
File: user_interface.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 4/4